SYSTEM
USER MANUAL

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System User Manual

Name	**ARJUN**
Company	**XAPA ENTERPRISES**
DEVICE	**BASE MODEL**

1. Introduction

The **BASE MODEL** developed by **XAPA Enterprises** is an intelligent monitoring and alert system designed for industrial and safety applications.

This board continuously monitors object presence or movement using compatible sensors (such as PNP proximity or photoelectric sensors). Based on sensor input, the system determines whether activity is present or absent.

If no movement or object detection is observed for a defined period, the system automatically triggers an alert mechanism. Notifications can be sent to up to **five predefined contacts**, ensuring timely awareness in critical situations.

The alert system operates with configurable time intervals, including:

- 5 minutes
- 15 minutes
- 30 minutes
- 60 mins
- 120 mins

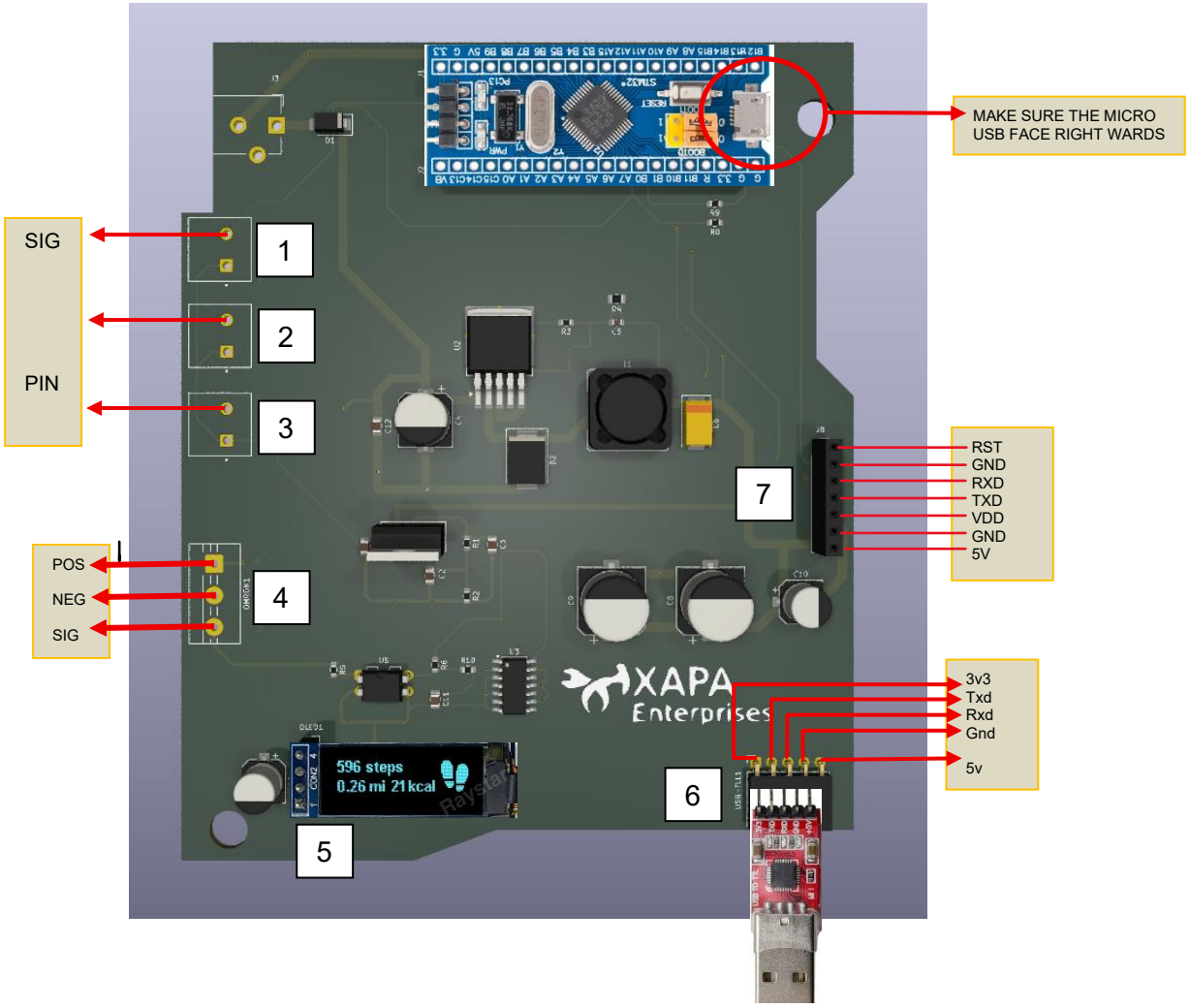
Users have full flexibility to **add, edit, or delete contact numbers at any time**, making the system adaptable to different operational requirements.

This product is suitable for applications such as machine monitoring, safety supervision, industrial automation, and remote alert systems.

1. System Setup

To get started with **BASE MODEL V2.0** follow these steps:

- **INSTALLATION [INSTALLATION** PROCESS]



TERMINAL 1: RESET in normal mode and DELETE number in configuration mode

TERMINAL 2: ADD/REPLACE any number in configuration mode, Switch any number in config mode

TERMINAL 3: ADD COUNT of scrap pieces/fault products on excel.

TERMINAL 4: PNP sensors of any 24v supply. Connect the Signal pin to Signal.

CONNECTOR 5: 0.96 inch Blue screen OLED 3.3v

CONNECTOR 6: USB – TLL for excel logging

CONNECTOR 7: SIM800L V2 for GSM with SIM SLOT

- **CONNECTION : [INSTRUCTIONS FOR CONNECTION]**

TERMINAL 1: Connect One terminal of NO PUSH BUTTON to POSITIVE and other to the SIGNAL pin.

TERMINAL 2: Connect One terminal of NO PUSH BUTTON to POSITIVE and other to the SIGNAL pin.

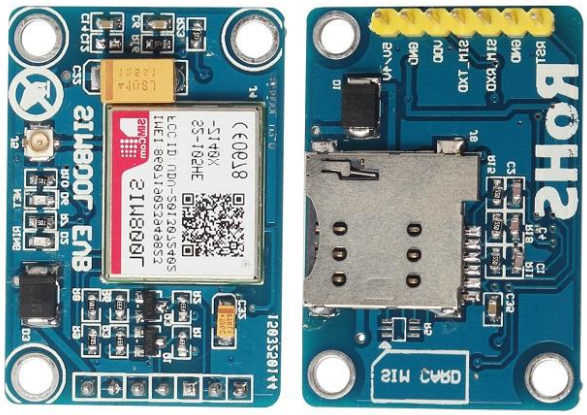
TERMINAL 3: Connect One terminal of NO PUSH BUTTON to POSITIVE and other to the SIGNAL pin.

TERMINAL 4: Connect POSITIVE of SENSOR to TERMINAL POSITIVE and GND to GND and SIGNAL pin to SIGNAL.

CONNECTOR 5 : Simply connect the OLED to the connector via dupont cable or any other cable, where 3.3v to POSITIVE, GND to GND.

CONNECTOR 6: Connect the TLL from right, where GND meets GND from right 2nd Connection

CONNECTOR 7 :



CONFIGURATION MODE : [INSTRUCTIONS FOR CALL/ADD/DELETE NUMBER FROM LOG]

ADD/REPLACE :

STEP 1: Press Button 2 to enter Config Mode. Keep pressing the button to switch between different profiles.

STEP 2: Select a profile and call the GSM number to register your number in the log if it is empty. If a number is already stored, call the same GSM number under the selected profile to replace the existing number.

STEP 3: To exit Config Mode, keep pressing Button 2 until it exits and returns to Normal Count Mode.

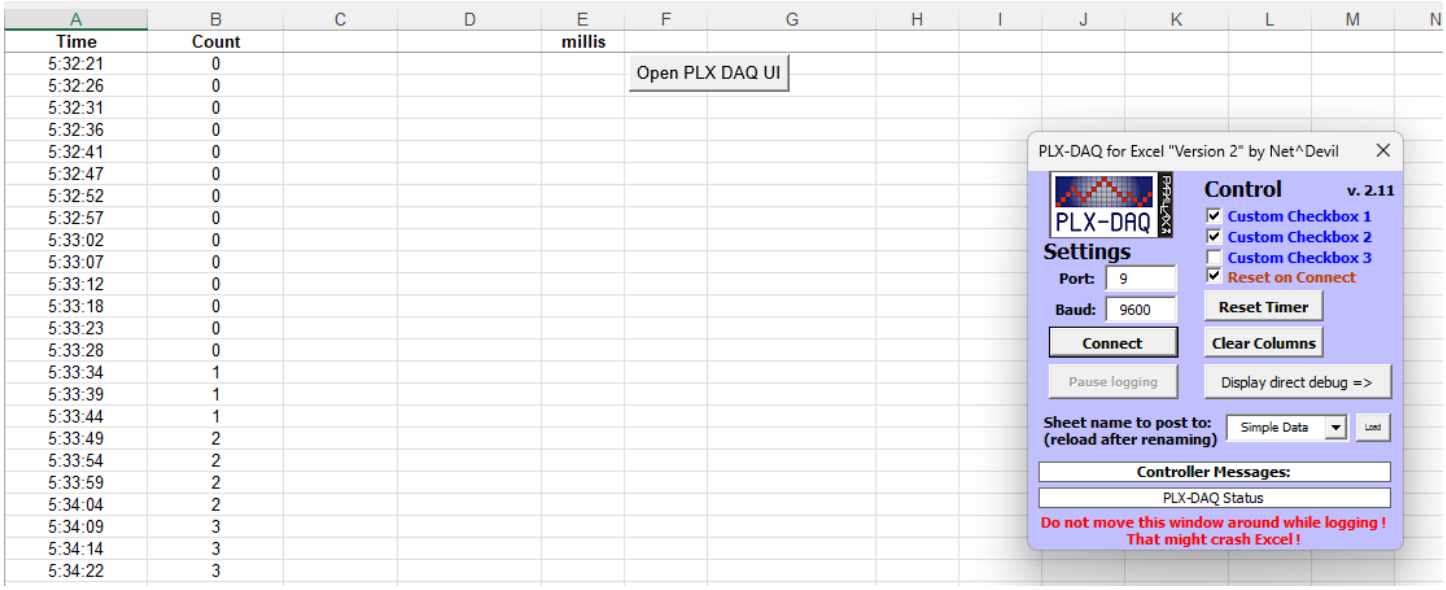
DELETE :

Select any profile that to be edited, then press the Button 1 to Delete the user number from the log.

EXCEL LOGGING [INSTRUCTION AND SOFTWARE INSTALLATION] :

STEP1 : Connect the **USB – TLL** port to the PC.

STEP2 : Install PLX- DAQ V2 application from browser https://forum.arduino.cc/t/plx-daq-version-2-now-with-64-bit-support-and-further-new-features/420628.



STEP 3: After Installation open the macro file, Set the baud rate to 9600 and connect.

Now the data's log in the excel sheet.

2. System Features

System Features

1. **Motion Detection and Monitoring**

   The system continuously monitors the surroundings using a motion sensor. When any movement is detected, the sensor triggers the microcontroller to initiate the alert process. This enables real-time detection of human presence or activity. (NOTE : ONLY PNP SENSOR WORKS NOT NPN (24V)). The count is visible in the display and EXCEL.

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2. **Automated Alert Notification System**

   Upon detecting no movement for predefined time intervals (5 minutes, 15 minutes, 30minutes, 60 mins , and 120mins), the system automatically sends alert messages to up to five registered contacts. This ensures timely communication and safety monitoring without manual intervention.

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3. **User Configurable Settings**

   Users can easily add, edit, or delete contact numbers and customize alert intervals as required. This flexibility allows the system to adapt to different user needs and usage scenarios.